
🔥 DDS – 30 MOST IMPORTANT QUESTIONS (UNIT-WISE)

> **Based on syllabus 303105201 & external exam trend**

🟦 UNIT 1: Introduction (Weightage: 10%)

→ 3 Questions

1. Define Data Structure. Explain its classification with examples.
2. Explain time complexity and space complexity. Analyze the complexity of a simple algorithm.
3. Explain pointers and dynamic memory allocation functions (`malloc`, `calloc`, `free`) with example.

🟦 UNIT 2: Stack, Recursion & Queue (Weightage: 15%)

→ 6 Questions

4. Explain stack operations using array representation with diagram.
5. Convert an infix expression into postfix expression and explain the algorithm.
6. Evaluate a postfix expression using stack with example.
7. Explain recursion with Tower of Hanoi problem.
8. Explain circular queue using array with diagram and operations.
9. Explain Deque and Priority Queue with suitable examples.

🟦 UNIT 3: Linked Lists (Weightage: 10%)

→ 4 Questions

10. Explain singly linked list representation and insertion operations with diagram.
11. Explain deletion operations in singly linked list.
12. Explain doubly linked list with advantages and disadvantages.
13. Explain circular linked list and header linked list with applications.

🟦 UNIT 4: Searching & Sorting (Weightage: 10%)

→ 4 Questions

14. Explain Linear Search and Binary Search with examples and complexity.
15. Explain Interpolation Search and compare it with Binary Search.
16. Explain Quick Sort algorithm and analyze its time complexity.
17. Perform Merge Sort on a given array and explain each step.

🟦 UNIT 5: Trees (Weightage: 10%)

→ 4 Questions

18. Define Binary Tree. Explain types and properties of binary trees.
19. Explain Binary Tree traversals (Inorder, Preorder, Postorder) with example.
20. Explain Binary Search Tree insertion and deletion operations.

21. Construct and evaluate an expression tree.

🟦 UNIT 6: AVL & Red-Black Trees (Weightage: 15%)

→ **5 Questions**

22. Explain AVL Tree and balance factor.
23. Explain LL, RR, LR and RL rotations in AVL Tree with diagrams.
24. Construct an AVL Tree for given set of values and show rotations.
25. Explain Red-Black Tree properties.
26. Compare AVL Tree and Red-Black Tree.

🟦 UNIT 7: Hashing (Weightage: 15%)

→ **2 Questions**

27. Explain hashing techniques and hash functions.
28. Explain collision and collision resolution techniques in hashing.

🟦 UNIT 8: Graphs (Weightage: 15%)

→ **2 Questions**

29. Explain graph representations: adjacency matrix and adjacency list.
30. Explain BFS and DFS traversal techniques with algorithms and examples.